

Arthropods

Phylum Arthropoda

Springtails

Class Collembola

Species About 8,100

Occurrence Grassland, leaf litter, soil

Springtails are named after their springy, fork-shaped tail, which is folded under the body and held there by a cliplike structure. When released, it flicks downward and catapults the insect into the air away from danger. Springtails swarm in damp places and water surfaces, feeding on organic plant material.

Silverfish

Order Thysanura

Species About 570

Occurrence Tree canopies, leaf litter, under stones, human dwellings

Silverfish differ from other insects in that they do not ever develop wings. Common in houses worldwide, they feed on starchy matter, such as flour, and various types of glue, such as wallpaper paste. Their scales detach easily, helping them to slip from the grasp of predators such as spiders.

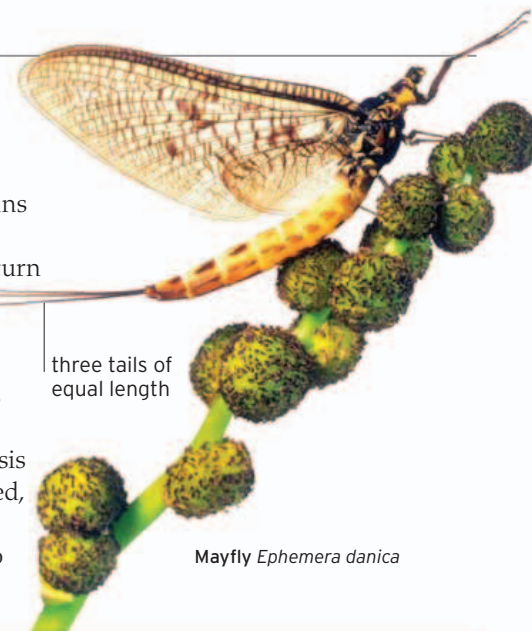
Mayflies

Order Ephemeroptera

Species About 3,000

Occurrence Fresh water, on vegetation

Renowned for their short lifespans as flying adults, mayflies live for months as aquatic larvae. They turn into nonfeeding adults for the brief final stage of their lives, when they mate, lay eggs, and die within a few hours. Mayflies were among the first insects to evolve this type of metamorphosis from a wingless larva to a winged, sexually mature adult, which is now typical of insects. They also have a winged subadult stage.

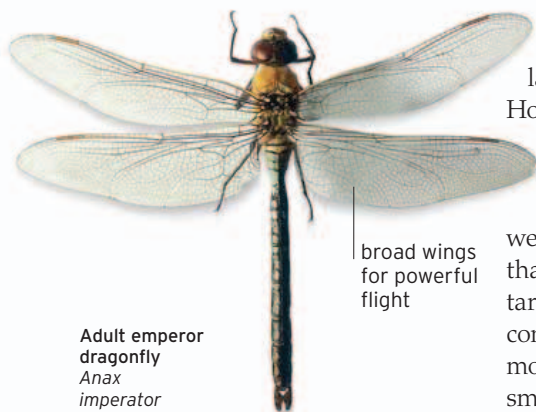


Damselflies and dragonflies

Order Odonata

Species About 5,600

Occurrence Still water, swamps, streams



The earliest of these insects evolved at about the same time as the mayflies, and they have many similar features, such as aquatic larvae that turn into flying adults. However, their larvae are voracious predators, and unlike mayflies, the adults are accomplished hunters that live for several weeks. Dragonflies are fast fliers that seize other insects on the wing, targeting them with their huge compound eyes. Damselflies are more delicate, fly weakly, and pick small insects off vegetation.

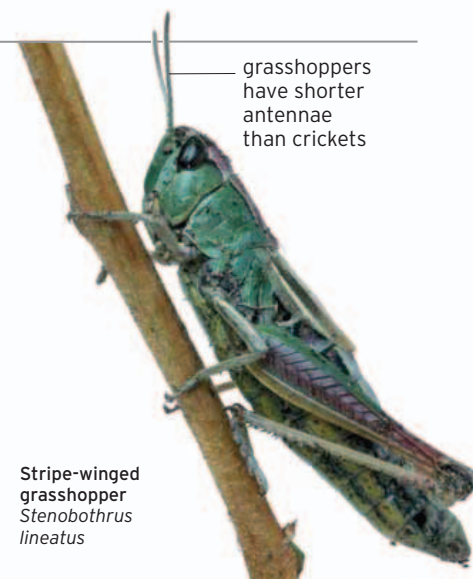
Crickets and grasshoppers

Order Orthoptera

Species Over 25,000

Occurrence On ground, on vegetation, damp soil, under logs and stones

These insects are recognizable by their long hindlegs, which grasshoppers in particular use to leap away from danger. They are also well known for their chirping songs. Most can fly, and the specialized grasshoppers known as locusts are notorious for flying in vast swarms that destroy farm crops. Crickets and grasshoppers do not undergo metamorphosis; the young are wingless miniatures of adults.



Stick and leaf insects

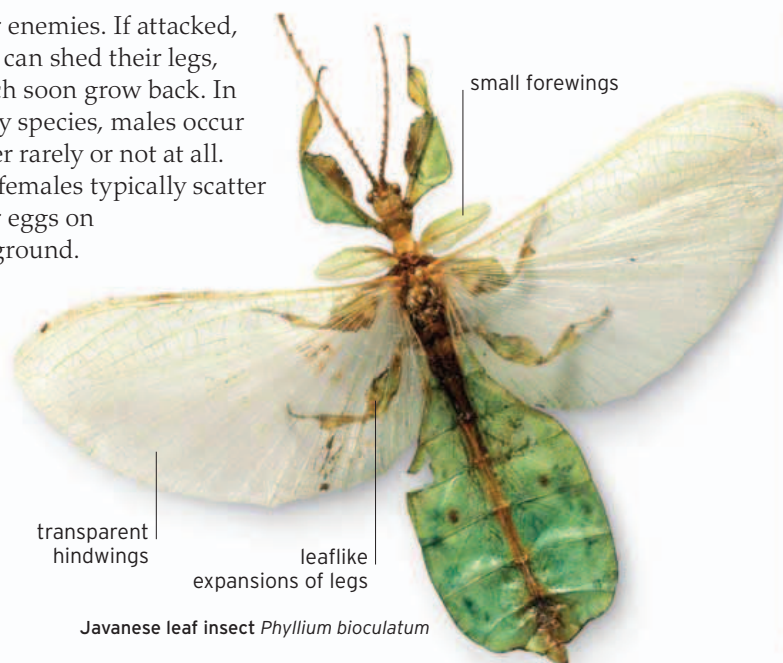
Order Phasmatodea

Species 2,450

Occurrence On vegetation, particularly foliage of trees and shrubs

Many insects are camouflaged to elude predators such as birds, but few can match the cryptic perfection of the stick and leaf insects. Shaped and colored to look almost exactly like twigs or leaves, the insects also mimic the way they grow and sway in the breeze. They use their chewing mouthparts to feed on plant foliage, moving slowly to avoid detection, but if this fails, some may defend themselves by squirting noxious chemicals at

their enemies. If attacked, they can shed their legs, which soon grow back. In many species, males occur either rarely or not at all. The females typically scatter their eggs on the ground.



Earwigs

Order Dermaptera

Species About 1,900

Occurrence Leaf litter, soil, under bark or in crevices

Most of the slender-bodied earwigs have a distinctive pair of sharp-pointed pincers at the end of their abdomen, which are used for defense, in courtship, and for grooming. They have thin, fan-shaped hindwings that they normally keep folded beneath their much shorter, hardened forewings. Earwigs eat a wide variety of plant and animal matter. The females are unusually maternal, guarding their eggs and licking them clean.